Innovative Lessons from Ladakh for Rest of the World

While the world wants to indigenise high-tech, Ladakh uses simple, local technologies to solve most of the problems. From solar-powered houses to mountain-to-mountain connectivity, Ladakh shows how simple methods can achieve huge milestones. This is a story not just about sustainable but regenerative development.

Living in Ladakh might seem like living in a remote part of the planet, especially when people feel cut-off from rest of the world for up to seven months in a year. With roads blocked due to heavy snow, landlocks exist in Ladakh till May or June every year, showing how remote the place is.

While problems vary from place to place and terrain to terrain, everyone is looking for simple and local solutions here to indigenise and solve them. Sonam Wangchuk from Ladakh believes, it would be interesting to know how the same science is being applied differently in Ladakh. He further explains how new digital technologies can be used to connect mountains, marking a huge milestone in network and connectivity.

Ladakh, located on the northernmost part of India, across the Himalayas, is quite a high and dry place with temperatures ranging from -35°C to +35°C from winter to summer. Situated on the Tibetan plateau—the rooftop of the world—this place is as close as it can get to outer space while still being on planet Earth. In terms of climate, Ladakh is considered a rarity, since when water freezes, problems like freezing, blockages, and bursting open of pipes happen. Yet, Ladakh is a thriving hub of innovation and problem solving.

Solar-powered houses

The launch of Students' Educational and Cultural Movement of Ladakh (SECMOL) has enabled children from the mountain range to have hands-on experience and learn things the practical way. Built entirely with mud from under the feet and powered by the sun over the head, this is one of the first schools that has had a zero energy off-grid campus.

> "We rescue it (the straw) and save it from polluting Punjab and Delhi. They become insulated blocks in Ladakh that are used to build the solar powered houses, thus cutting pollution in both Punjab and Ladakh."
>
> —***Sonam Wangchuk***

With abundant sunshine available almost up to 300 days a year, solar-heated mud houses have been built in Ladakh. Despite their low cost of construction, these houses are efficient enough to provide +18°C environment in a -20°C winter, with nothing but the sun to help.

Himalayan Institute of Alternatives, Ladakh is also working on some local solutions. "One of the interesting experiments is where we've built

Figure 1: The solar-powered and solar-heated SECMOL